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Detecting Trust Calibration Traits in AI with EEG Signals for Speech Deception

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Abstract. This paper investigates the effectiveness of frequency-based electroencephalogram (EEG) measures in capturing self-reported trust and trust miscalibration in artificial intelligence (AI) systems used as decision support tools. It examines a collaborative human-AI decision-making task where 50 human users interacted with an explainable speech-based AI system to detect deceptive speech. Correlation analysis indicates that Alpha and Theta power values of EEG measured from central, frontal, and parietal regions depict negative correlation with self-reported trust, and Delta power values from frontal regions are negatively correlated with self-reported trust. A machine learning model using EEG power measures to estimate self-reported trust depicted a Spearman's correlation value $\rho = 0.63$ (p <0.01) between predicted and actual self-reported trust. Additionally, the Beta-band-power values from left frontal and left central areas are higher during trust calibration compared to under-trust. The Gamma-band-power levels are also higher during trust-calibration compared to over-trust. Machine learning models based on these EEG measures predict different trust dimensions (i.e., over-trust, under-trust, trust-calibration) with moderate macro-F1 scores (i.e., 53-55%). Finally, the most effective trust assessment models leverage data from all considered brain regions – frontal, central, and parietal – achieving similar performance compared to models using central regions only and outperforming models using frontal or parietal regions alone.

Keywords. EEG signals, trust in AI, deceptive speech, trust calibration, human-AI

1. Introduction

Artificial intelligence (AI) has shown remarkable success across a wide variety of fields including healthcare [31, 57], the military [26], security [20, 24, 25], and management [27], among others. However, in certain human-centered decision-making tasks, such as deception detection or emotion recognition, AI systems often struggle to achieve satisfactory performance due to the subjective nature of these tasks, which rely heavily on human intuition and cognitive abilities. On the other hand, humans may also face challenges in these tasks, as they involve subtle variations in observable cues, such as speech,

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facial expressions, or gestures, that are often difficult to perceive by humans, but can be more readily identified by AI systems. Detecting deception, for instance, is a notoriously difficult task for which neither AI nor humans have yet achieved consistently reliable results. For example, when distinguishing between deceptive and truthful opinions regarding abortion, ML models using lexicon-based features had a 67% accuracy, compared to 52% accuracy for human judges [59]. For classifying between deceptive and truthful speech, the best F1 score achieved by a speech-based ML model was 72.77%, while human judges achieved a 46% on the same task [51]. Therefore, a collaborative effort between humans and AI can potentially achieve significant performance improvements in these challenging areas.

Trust calibration is a fundamental aspect of effective human-AI collaboration [43]. Lee & See [48] defined trust in automation as "*the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability*." For human-AI collaboration to be successful and outperform either agent working alone, users need to understand when to place their trust in AI. Without this understanding, user trust can either exceed the system's capabilities leading to over-trust, or fall short, leading to under-trust and under-utilization of the system. Existing research identifies three trust-calibration dimensions [48]: (1) Calibrated trust: users appropriately trust the correct AI or appropriately distrust the incorrect AI; (2) Under-trust: users fail to trust the correct AI; and (3) Over-trust: users place trust in the incorrect AI. Under-trust in AI may result in suboptimal decision making and task performance [79], while over-trust in AI may yield faulty decisions with severe consequences (e.g., crash of Air320, 1999 [34]). EEG power measures, such as Beta waves, could be indicative of trust calibration, which entails logical reasoning and conscious evaluation, since Beta brain activity is associated with active decision-making, focus, and critical thinking [19]. Measuring a human user's trust in the AI system in real-time can enhance human-AI collaboration by allowing the AI system to adapt its strategies based on the user's trust level [53], potentially increasing user satisfaction and mitigating critical errors.

Self-report measures are the most straightforward way to gauge trust [32, 80], but they lack the ability to provide continuous measurements. Researchers have found that eye gaze behaviors [28, 32], heart rate changes [58, 74], electrodermal activities [5, 40], neural measures [5, 81], and facial expressions [45, 70] can be used effectively to measure trust in real time. Recent studies show that electroencephalogram (EEG) signals have emerged as promising solutions to develop real-time trust assessment models [5, 13, 81]. EEG signals could explain particular cognitive state details that are essential to trust [10, 35, 55, 63, 83]. Furthermore, EEG's high temporal resolution makes it feasible to record abrupt shifts in brain activity over brief periods of time, which is essential for dynamic trust monitoring in real-time. Zhang *et al.* [81] found that Beta waves are negatively correlated with self-reported trust in automated vehicles, possibly because elevated Beta power could be linked to increased cognitive processing, and drivers are more alert to monitoring the status of automated vehicles when they do not trust the system. Their analysis suggests that the frontal and parietal regions have the greatest potential to develop trust assessment models. Oh *et al* [60] demonstrated that Alpha, Beta and Gamma waves are correlated to trust, while other studies found that [73] Delta and Gamma waves have the most significant correlations with trust levels. Some studies [73] found that the frontal and occipital regions of the brain correlate with human trust in AI, most likely because these regions are responsible for executing cognitive functions [15], and processing visual information [4] respectively. Also, Akash *et al.* [5] found that time-domain EEG features from central region of the brain are among the top features based on the ReliefF weights [42, 44] to predict trust in intelligent systems. EEG signals have served as input to machine learning models to build continuous trust assessment models with promising results. Zhang *et al.* [81] used time and frequency domain EEG features to develop a driver trust assessment model, achieving an accuracy of 88.44% and an F1 score of 78.31%. Other researchers achieved similar accuracy on a trust sensor model using EEG features (e.g., $73.13 \pm 0.01\%$ [5], $71.57 \pm 0.05\%$ [33]). Although researchers have found that EEG features have the potential to detect trust in automation, there has not been any work on detecting properly calibrated trust against mis-calibrated trust (e.g., over-trust, under-trust) using EEG features, especially in a subjective task like deceptive speech detection, which is the main focus of this paper.

Here, we investigate the effectiveness of EEG measures in approximating self-reported measures of trust in AI, as well as detecting instances of user trust mis-calibration, including over-trust and under-trust, in AI. We aim to answer the following research questions:

- **RQ1:** What is the relation between EEG signals and self-reported trust?
- **RQ2:** Can EEG signals distinguish between instances of proper trust calibration and trust mis-calibration, including over-trust and under-trust, in AI?
- **RQ3:** Which brain region can be most effectively used to develop trust assessment models?

Our results suggest that Alpha and Theta power values show significant negative correlations with self-reported trust when these EEG measures are extracted from frontal (F3, F4), central (C3, C4), and left parietal (P3) regions. Apart from that, Delta power values from frontal regions (F3, F4) are negatively correlated with self-reported trust. Our results also indicate that Beta-band-power values from left frontal and left central regions are higher when the users properly calibrate their trust compared to when they under-trust AI. Furthermore, we found that Gamma band-power values from left parietal region are significantly higher when users properly calibrate their trust in AI than when they over-trust AI. When training machine learning models to automatically estimate self-reported trust based on the aforementioned EEG measures, we obtain a Spearman's correlation value of $\rho = 0.63$ ($p <0.01$) between the actual and predicted self-reported trust. For the task of automatic classification between trust calibration and over-trust, as well as proper-trust and under-trust, using EEG features, we further obtained 53-55% macro F1-score. Finally, our results indicate that EEG measures based on the central region are the most effective in serving as input to machine learning models of self-reported trust, followed by measures obtained by the frontal and parietal regions. However, we did not find significant differences among the three regions in terms of their effectiveness in classifying between trust calibration and trust mis-calibration dimensions.

2. Prior Work

EEG signals record post-neuronal electrical activity in the brain, providing insights into neurocognitive states and processes, such as attention, memory, perception, and decision making [23, 72]. Empirical evidence from prior research indicates that EEG measures correlate with human attention state, consciousness, decision-making, and cognitive load [6, 7, 22, 29, 52]. There are five major brain waves, each distinguished by its frequency band: Delta (0–4 Hz), Theta (4–8 Hz), Alpha (8–12 Hz), Beta (12–30 Hz), and

Gamma (30–45 Hz). These brain waves are related to specific brain activities. For example, the Alpha band-power decreases significantly with an increase in attention, compared to the relaxed state, in frontal (F3 and F4) and occipital regions (O1 and O2) [68]. Beta-band activity is related to consciousness, concentration, cognitive processes, and critical thinking [8,19]. Similarly, gamma waves can indicate cognitive information, such as reasoning and judgment [38]. On the other hand, Delta and Theta are the lowest frequency waves which are primarily related to sleep [14]. Since trust often relies on cognitive processes such as consciousness, attention, and critical thinking, these findings indicate that it may be possible to capture human trust based on EEG measures. Different areas of the brain are responsible for distinct functions. Frontal lobes play a key role in cognitive processing, memory, planning, and attention [15], while the occipital lobes are responsible for visual processing [65]. On the other hand, the central and temporal lobes are linked to mental workload [36]. Finally, prior work indicates that the parietal lobe is related to spatial attention [9].

Recently, significant research efforts have been directed toward finding potential associations between human trust and EEG features. Oh *et al.* [60] found that when comparing trust to distrust conditions, the power of Alpha and beta waves was higher during trust, while the power of gamma waves was higher in distrust. In a study by Wang *et al.* [73], 10 human users collaborated with an AI system to play an investment game under different trust scenarios. Based on linear mixed effect analysis, the frontal and occipital brain areas were identified as being the most sensitive to changes in trustworthiness of the AI agent. In addition, they also found that Delta and gamma waves were significantly correlated with agent's trustworthiness.

Other efforts have focused on training machine learning models based on EEG measures to automatically estimate human trust. Zhang *et al.* [81] proposed a Light Gradient Boosting Machine (LightGBM) to classify between low, medium, and high levels of self-reported trust in automated vehicles using the EEG features. The proposed model achieved an accuracy of 88.44% and F1 score of 78.31%. Their results further suggest that models using the features from all brain regions show superior performance than that of individual ones. In addition, models based on frontal and parietal brain regions outperform the models using only features from the occipital region. Their findings indicate that higher beta power waves from CP2, and CP5 indicate a lower level of trust. Akash *et al.* [5] measured real-time trust using EEG and galvanic skin response features. The users were asked to drive a car equipped with an obstacle sensor that will alert them if an obstacle is ahead. EEG signals collected during the study were divided into 1-second epochs with a 50% overlap. Following that, each epoch was labeled as either trust or distrust. A Quadratic Discriminant Analysis classifier achieved an average accuracy of 73.13% in this binary classification task. Furthermore, beta-band-power values from the left parietal lobe (P3) were the top features selected for the trust classification models. In a similar line of work, Hu *et al.* [33] achieved an average accuracy of 71.57% in classifying trust in two levels (i.e., trust or distrust) in automated vehicles using a combination of EEG features, galvanic skin response features, and task response time. In addition, high beta-band waves from the right hemisphere and the parietal lobe (C4, P4, POz) and mid beta-band waves from the left hemisphere (C3) respond strongly to discriminate between accurate and faulty trials.

In relation to prior work, this paper offers the following contributions: (1) It extends previous research [5, 33, 81] that explored associations between EEG measures and self-reported trust in automation by investigating real-time models of trust calibration and mis-calibration based on EEG features. To our knowledge, this represents the first study in this area; and (2) While prior studies investigating EEG correlates of trust have primarily focused on trust in automated vehicles [5, 33, 81], this paper examines a decision-making task involving collaboration between a human user and an explainable AI system for detecting deceptive speech. The focal task is considerably more challenging and cognitively demanding than those considered in previous work (e.g., obstacle detection [5, 33], taking manual control from automated vehicle [81]).

3. Design of explainable AI used as decision support tool

We built an explainable AI model that was used as the decision support tool for the participants in the user study. The model was trained based on deceptive and non-deceptive speech data from the Columbia X-Cultural Deception (CXD) Corpus [51]. The CXD Corpus included a total of 340 participants who had to answer a set of 24 biographical questions. Participants were instructed to provide the true answer to a random half of the questions and a false answer to the rest. These responses were used as ground truth. In total, 4373 question-answer pairs were used to build the explainable AI model.

We used a total of 7 linguistic and acoustic features depicted as useful indicators of deception in prior work [17, 18, 51, 77]. The linguistic features included the percentage of negations (e.g., no, not) and total number of function words (e.g., it, to) extracted by the Linguistic Inquiry and Word Count (LIWC) toolbox [18], mean pleasantness in the response extracted by the Dictionary of Affect in Language (DAL) [77], creativity [17], and number of filled pauses [17]. Acoustic features included the maximum speech amplitude and maximum pitch within the response extracted using PRAAT [11, 51].

We used the explainable boosting machine (EBM) [54] to classify deceptive speech. EBM estimates the deception outcome y based on a set of features x_j as follows:

$$\mathscr{E}[y] = \beta_0 + \sum_j f_j(x_j) \tag{1}$$

where $\mathscr{E}$ is the expected value of the outcome, β_0 is the intercept, and f_j is a feature function that shows the contribution of each feature x_j to the model's output y. The EBM was trained on 75% of the CXD data and the remaining 25% data was employed as the test set. By varying the hyper-parameters of the EBM, we built a high performing (HP) AI and low performing (LP) AI. The HP model achieved 59% F1-score and LP model achieved 36% F1-score. We randomly selected 40 samples from the test set that were shown to the participants in our user study. The EBM model provides a global explanation graph, which shows the correlation between each feature and speech deception based on all the data. Additionally, the EBM also gives a local explanation graph on the sample-level which explains the contribution of each feature in estimating the deception for each audio sample.

4. User Study Design

4.1. Participant Recruitment

We recruited 52 participants (22.63 ± 6.62 years; 40 female, 11 male, and 1 other) via mailing lists at Texas A&M University. Data included 10 Asian participants, 3 Black, 8 Hispanic/Latino, 1 Native Hawaiian, 27 White, and 3 related to people of more than one racial or ethnic group. In terms of education level, 42 participants were undergraduate students, 8 were graduate students, and 2 participants were post-graduate students. Due to a recording error, we excluded the data from 2 participants.

4.2. Study Protocol

During the day of the study, a member of the research team explained the study protocol to the participants. Each participant viewed a mini-tutorial which explained the goal of the task. From the total set of participants, 30 were randomly assigned to the HP AI condition and 22 were assigned to the LP AI condition. Participants interacted with the assigned AI via a user interface. First, participants reviewed the global explanation graphs in order to understand the overall association between each feature and the probability of deception provided by the EBM model. Following this, for each sample they read the interview question and listened to the audio response to the question for that sample. Then participants viewed the local explanation graph for the corresponding audio sample along with the AI decision for that sample. Finally, participants provided their decision about whether the audio sample response was deceptive or not. Participants were instructed that their own decision does not necessarily need to be aligned with the AI decision. This was repeated for all 40 audio samples that had been selected for this user study. After providing their decisions on 8 consecutive audio samples (i.e., one batch), participants were asked to rate their perceived trust in AI on a 1-5 Likert scale (1: Low Trust, 5: High Trust), which served as a measure of self-reported trust. Each participant was compensated with a $50 Amazon gift card.

During the study, the participants were equipped with an 8-channel Enobio headset [1] that was used to collect their EEG signals, and NIC2 [2] software was used to control the headset from the computer. The EEG signals were sampled at $500Hz$ frequency. The electrode placements followed the international 10-10 system [12]. We applied highly conductive electrode gel [3] on the contact surface between the Ag/AgCl electrode and the scalp, in order to decrease the impedance. To ensure optimal signal quality, the contact impedance between the electrodes and the participant's skin was kept below 10 kΩ. We collected signals from 8 channels, including $F_3, F_4, C_3, C_4, P_3, P_4, F_7, F_8$, with the right earlobe taken as a reference, as depicted in Figure 1(a). Due to the observed measurement noise, we did not use features from F7 and F8 in our analysis.

As we did not find any significant difference between user trust and user performance between HP and LP conditions, the subsequent analysis reported in this paper includes the combined data from both LP and HP conditions.

5. Methods

5.1. EEG measures

We used the default line noise filter from NIC2 [2] to remove the main line artifacts from EEG data at 60Hz. We extracted features from the five basic frequency bands, that include Delta (0-4 Hz), Theta (4-8 Hz), Alpha (8-12 Hz), Beta (12-30 Hz), and Gamma (30-45 Hz) across six channels (i.e., F3, F4, C3, C4, P3, P4) [62,67]. We defined the reaction time from the moment the user stopped listening to an audio sample until they made a choice. We then used the EEG signals during the reaction time interval to predict the deception for that sample. We divided each reaction time interval into 1 sec epochs with 50% overlap and measured the average band-power values via Welch's algorithm [75] from each epoch for each of the corresponding frequency bands, resulting in a total of 30 EEG measures (i.e., 6 channels $\times 5$ frequency bands) per audio sample that each participant reviewed. For outlier detection, we used inter-quartile range (IQR) method. We define the IQR distance for any band-power as the distance between the third quartile (i.e., Q3) and the first quartile (i.e., Q1) of the corresponding band-power

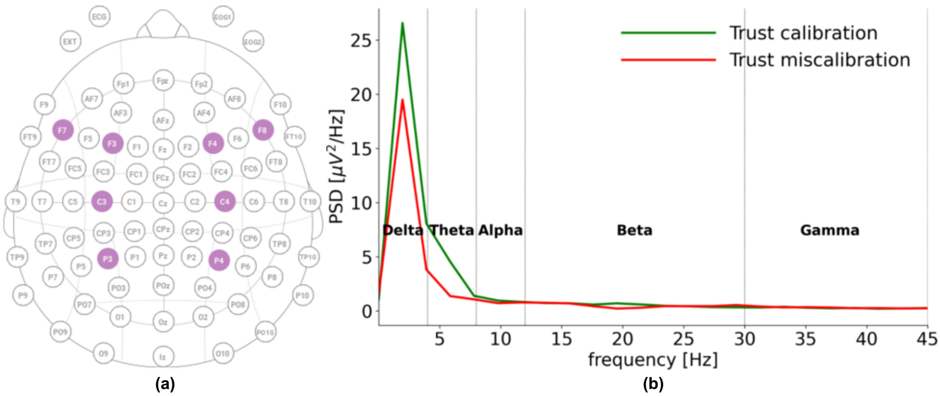

Figure 1. (a) Placement of the electroencephalogram (EEG) electrodes in the Enobio 8 headset [1] according to international 10-10 system [12]. The used electrodes in the user study are marked as purple. This image was generated with the NIC2 software [2]; (b) A sample power spectrum density (PSD) plot for the EEG signals from F3 lobe where x axis denotes the frequency and the y axis presents the PSD values. Here, we have shown the PSD plot corresponding to a trust calibration (i.e., green line) and a mis-calibration case (i.e., red line).

values [76]. In this method, if the band-power value lies more than 1.5 IQR points below Q1 or more than 1.5 IQR points above Q3, we remove that epoch from the analysis. A sample power spectrum density (PSD) plot highlighting our frequency range is shown in Figure 1(b). In this figure, we used the extracted EEG signals from the F3 frontal lobe for a sample where a user properly calibrated their trust in AI and for another sample where the same user did not calibrate their trust in AI.

5.2. *Association between self-reported trust and EEG signals*

We investigate potential associations between neural measures captured via EEG and self-reported user trust in AI. We computed the Pearson's correlation between each EEG feature, averaged over all batches (i.e., one batch containing 8 consecutive audio samples, as described in Section 4.2), and self-reported trust in AI. Next, we built support vector regression (SVR) models with radial basis kernel using EEG features to predict the self-reported trust. We normalized all the features before training the model. The EEG measures include 5 basic band-power measures (Alpha, Beta, Theta, Gamma, Delta) across 6 channels, resulting in 30 (i.e., 6 channels $\times 5$ frequency bands) features, as discussed in Section 5.1. We obtained results using a leave-one-participant-out cross validation framework, according to which each fold included data from a single participant in the test set and the remaining participants in the train set, a procedure which was repeated as many times as the number of participants. Given the large inter-individual variability in EEG signals [39], we have further taken 25% samples from each target participant into the training set to reduce the individual bias in the training. Furthermore, we assigned more weight to the 25% instances from the test participant in the training set so that the model can capture the individual variation of EEG signals in the target participant. Following that, we computed the Spearman's correlation *rho* between predicted and actual self-reported trust in AI as the performance metric.

5.3. *Association between trust-calibration dimensions and EEG signals*

We explored the extent to which EEG measures present differences between properly calibrated trust and mis-calibrated trust, including over-trust and under-trust. Since trust calibration is done on the sample level, we used sample-level trust for this analysis. We

Table 1. Pearson's correlation (N =246) between self-reported trust in AI and electroencephalogram (EEG) measures.

Channel	Alpha	Beta	Gamma	Theta	Delta
F_3	-0.366^{***}	-0.024	-0.095	-0.301^{***}	-0.194^{**}
F_4	-0.179^{**}	0.026	-0.026	-0.228^{**}	-0.153^{*}
C_3	-0.205^{**}	0.073	-0.013	-0.172^{**}	0.007
C_4	-0.141^{*}	0.016	-0.033	-0.198^{**}	-0.096
P_3	-0.194^{**}	0.037	-0.037	-0.160^{*}	-0.041
P_4	-0.068	0.030	-0.042	-0.121	-0.016

***,**, *: $p < 0.001$, $p < 0.01$, $p < 0.05$

took average EEG measures for different trust dimensions per person and then conducted a paired t-test to see if EEG signals for trust mis-calibrated regions (e.g., over-trust, under-trust) significantly differ from trust-calibrated regions. In addition, we built class weighted support vector classifier (SVC) models with radial basis kernel using EEG features to classify different trust dimensions. We built two types of models: one model to classify properly calibrated trust against over-trust, and the other model to classify properly calibrated trust against under-trust. We used the same normalization procedure, features, and training-testing strategies as described in Section 5.2. We report macro F1 score, macro recall, sensitivity, and specificity of the classification models. We have shown model results where positive class is properly calibrated trust and negative class is under-trust. Apart from that, we have also demonstrated model results where positive class is properly calibrated and negative class is over-trust.

5.4. *Effectiveness of various brain regions in capturing trust in AI*

We developed separate SVR models based on EEG features from individual brain regions, namely frontal, central, and parietal. Also, we built class-weighted SVC models to distinguish between properly calibrated trust and mis-calibrated trust, including both over-trust and under-trust. No sampling strategy was employed in these comparisons, as our primary goal was to analyze which brain region is most effective in assessing trust.

6. Results

6.1. *Correlation between self-reported trust and EEG measures*

The Pearson's correlation between self-reported trust and EEG measures is reported in Table 1. Alpha and Theta power values from frontal (i.e., F3, F4), central (i.e., C3, C4), and left parietal (i.e., P3) brain regions have significant negative correlation with self-reported trust. Also, we observed a negative trend between Theta power values from the right parietal brain region (i.e., P4) and self-reported trust, although the effect is not statistically significant (i.e., $Pearson's(r)$=-0.121, p=0.059). Apart from that, Delta power values from frontal brain regions (i.e., F3, F4) have significant negative correlation with self-reported trust.

6.2. *Self-reported trust assessment model results*

We obtain a Spearman's correlation value $\rho = 0.21$ ($p < 0.01$) using leave-one-participant-out cross-validation for estimating self-reported trust based on the EEG measures without using any data from the test participant in the training set (Table 2). When using a randomly selected subset of 25% samples from test participant in the training data, we obtain an increase in Spearman's correlation to $\rho = 0.43$ ($p < 0.01$). This improvement is not statistically significant compared to the performance without includ-

Table 2. Spearman's correlation (N =246) between actual and predicted self-reported trust values using support vector regression (SVR) models.

Sampling	Sample weight	Spearman's correlation (ρ)
X	X	0.21^{**}
✓	1×	0.43^{**}
✓	2×	0.50^{**}
✓	5×	0.60^{**}
✓	10×	$\mathbf{0.63}^{**}$
✓	20×	0.63^{**}

$^{**}, ^{*}$: $p < 0.01$, $p < 0.05$

ing data from the target speaker in the train, as assessed by a one-tailed t-test (i.e., $t(34)$=0.116, p=0.45). Assigning more weight to the random 25% samples from the test participant that were used in the training set further improves performance. We obtain a maximum Spearman's correlation of $\rho = 0.63$ ($p < 0.01$) using 10× more weight in the samples from the test participant during training. This result demonstrates an improving trend, compared to the one obtained without including data from the target participant in the test, though the effect is not statistically significant based on a one-tailed paired t-test (i.e., $t(34)$=1.20, p=0.11).

6.3. *Difference of EEG signals across trust calibration dimensions*

Beta wave power values from left frontal (i.e., F3) and left central (i.e., C3) regions are significantly higher during audio samples in which the participant presents properly calibrated trust in the AI compared to under-trust (i.e., $t(49)$=2.01, p=0.03 for F3; $t(49)$=1.76, p=0.04 for C3). In addition, Gamma power values in the left parietal region (i.e., P3) are significantly higher in properly calibrated region compared to over-trust region (i.e., $t(49)$=1.67, p=0.04 for P3). The rest of the t-tests did not show statistically significant results for any EEG band-power values and any channel.

6.4. *Real-time trust-calibration assessment model results*

Results indicate a macro F1-score of 0.46 using leave-one-participant-out cross-validation in classifying differences between properly calibrated trust against under-trust (Table 3). Using a random set of 25% samples from the test participant for training and adding more weight to those samples resulted in significant performance improvements. The maximum macro F1-score of 0.53 was obtained when classifying between properly calibrated trust against under-trust and assigning 10× more weight to samples from the test participant during training. This improvement in performance is statistically significant compared to when data from the target participant is excluded, based on a one-tailed paired t-test (i.e., $t(49)$=3.36. p=0.0007). A macro F1-score of 0.49 was obtained when classifying between properly calibrated trust against over-trust without using any samples from a test participant in training. The same classification task resulted in a maximum macro F1-score of 0.55 when assigning 10× more weight to 25% randomly selected samples from the test participant, a performance improvement that is also statistically significant based on a one-tailed paired t-test (i.e., $t(49)$=3.81, p=0.0001).

6.5. *Trust assessment model results using different brain regions*

We obtain a Spearman's correlation value of $\rho = 0.10$ (p=0.12), $\rho = 0.193$ (p=0.002), and $\rho = 0.072$ (p=0.255) between self-reported and predicted trust using the EGG features from frontal, central, and parietal regions respectively. Based on a two-sided paired t-test, we found no significant difference in performance for these models when compar-

Table 3. Classification model results where the positive class is properly calibrated trust and the negative class is under-trust or over-trust. The under-trust classification model is based on 1486 samples (974 positive, 512 negative). The over-trust classification model is based on 1458 samples (974 positive, 484 negative).

Sampling	**Sample weight**	**Under-trust classification model**				**Over-trust classification model**			
		Macro F1	**Macro Recall**	**Sensi-tivity**	**Speci-ficity**	**Macro F1**	**Macro Recall**	**Sensi-tivity**	**Speci-ficity**
✗	✗	0.46	0.48	0.45	0.51	0.49	0.49	0.54	0.45
✓	1×	0.49	0.51	0.45	0.58	0.50	0.52	0.52	0.51
✓	2×	0.50	0.53	0.47	0.59	0.51	0.52	0.51	0.53
✓	5×	0.53	0.55	0.53	0.58	0.54	0.56	0.53	0.58
✓	10×	**0.53**	0.55	0.53	0.56	**0.55**	0.56	0.58	0.55
✓	20×	0.53	0.54	0.58	0.50	0.55	0.55	0.63	0.48

ing between frontal and central regions (i.e., $t(42)$=−1.77, p=0.08), frontal and parietal regions (i.e., $t(42)$=−0.72, p=0.48), and central and parietal regions (i.e., $t(42)$=0.98, p=0.33). Although we obtain the best performance using the whole brain regions together, resulting in a Spearman's correlation of $\rho = 0.21$ (p=0.001), the performance difference between the central brain regions compared to all brain regions is not statistically significant (i.e., $t(42)$=−0.36, p=0.72). Therefore, we can use the central region only for assessing self-reported trust without significant loss in performance compared to using the entire brain. Also, we obtain the same macro F1-score of 0.45 based on frontal, central, and parietal regions when classifying between properly calibrated trust against under-trust. When classifying between properly calibrated trust against over-trust, the corresponding models achieve a macro F1-score of 0.49, 0.47, and 0.47 based on the frontal, central, and parietal regions, respectively. Based on a two-sided paired t-test, we found no significant difference in the performance of these models when comparing between frontal and central regions (i.e., $t(49)$=0.023, $p = 0.981$; $t(49)$=1.13, p=0.26 for under-trust and over-trust classification models, respectively), frontal and parietal regions (i.e., $t(49)$=0.27, p=0.78; $t(49)$=0.81, p=0.42 for under-trust and over-trust classification models, respectively); and central and parietal regions (i.e., $t(49)$=0.328, $p = 0.74$; $t(49)$=−0.40, p=0.69 for under-trust and over-trust classification models, respectively). Therefore, we obtain no significant differences in classification models using different brain regions to classify properly calibrated trust against mis-calibrated trust.

7. Discussion

In response to **RQ1**, our results suggest that Alpha and Theta power values from frontal (i.e., F3, F4), central (i.e., C3, C4) and left parietal (i.e., P3) regions show significant negative correlation with self-reported trust. Apart from that, Delta power values from frontal brain regions (i.e., F3, F4) also depict significant negative correlation with self-reported trust (See Section 5.2, 6.1, Table 1). Theta waves appear in sleep [64]. Both Theta and Alpha waves indicate relaxation [64, 78]. Previous research also illustrates that Delta waves indicate deep sleep and suspend external awareness [49, 64]. These suggest that when a user depicts decreased trust in AI, their cognitive engagement with the AI system may decrease, leading to reduced mental effort in evaluating the AI output and explanation, thus potentially being more relaxed, resulting in an increase in the corresponding Alpha, Theta and Delta values of the EEG measures. When predicting self-reported trust based on the frequency domain EEG measures, we obtain a maximum Spearman's correlation of ρ=0.63 (p<0.01) between the actual and predicted self-reported trust including 25% random samples from the test participant in training with 10× more weight (See Section 5.2, 6.2, Table 2). Interestingly, our results demonstrate that we can improve performance using random samples from a test participant in training compared to not including any samples from the target participant (i.e., Spearman's correlation value ρ=0.43 (p<0.01) compared to ρ=0.21(p<0.01)), though this improvement is not statistically significant. However, we found that assigning 10× more weight to random samples from the test participant during training shows a superior performance trend, i.e., Spearman's correlation of ρ=0.63 (p<0.01) compared to ρ=0.21 (p<0.01), although the effect is not statistically significant (i.e., $t(34)$=1.20, p=0.11). The improving trend in prediction accuracy when random samples from the test participant are included in training suggests that models greatly benefit from individual-specific EEG patterns, enabling more accurate trust predictions. Prior work used EEG measures to build classification models to predict trust in automation, getting an average accuracy of 71.57% [33], 73.13% [5], 88.44% [81]. However, none of the previous work built regression models which can predict the level of trust, rather than the presence of trust in AI.

In response to **RQ2**, we analyzed which EEG signals can differentiate properly calibrated trust regions from mis-calibrated trust, including under-trust and over-trust. We found that Beta power values from the left frontal region, F3, and the left central brain region, C3, are significantly higher when users properly calibrate their trust in AI compared to when users under-trust the AI (See Section 5.3, 6.3). Our results also indicate that Gamma power values in the left parietal brain region, P3, are significantly higher when users calibrate their trust in AI appropriately, as compared to when they over-trust the AI. Prior work [64] suggests that Beta brain activities are linked to active decision-making and intense cognitive function. Beta waves are commonly observed in an "awaken" state and appear when one is focused, making decisions, and using judgment [19]. Having the right amount of Beta waves allows individuals to focus on critical thinking [19]. Collectively, these could suggest that Beta waves are higher when users properly calibrate their trust in the AI, potentially because users engage in conscious thought and logical thinking during proper trust calibration. In addition, prior findings [30, 46, 50, 69] illustrate that Gamma band waves play a crucial part in the processes underlying states of consciousness. Gamma waves appear in high level cognition and attention [19, 37, 41]. These findings indicate that Gamma waves occur when users properly calibrate their trust, because they pay increased attention and are more aware of the decision-making process compared to when they do not calibrate their trust in the AI. In addition, machine learning models based on the EEG measures achieved a macro F1 score of 0.53 to classify properly calibrated trust against over-trust using 10× more weight to 25% randomly selected test samples in training (See Section 5.3, 6.4, Table 3). Apart from this, we obtain a macro F1 score of 0.55 from the classification model to predict properly calibrated trust against over-trust. These performance improvements are statistically significant, as compared to when we do not use data specific to a target participant, indicating again the importance of personalized data and weighted training for individual adaptation.

In response to **RQ3**, our results indicate that central regions can effectively predict self-reported trust (See Section 5.4, 6.5). Prior work [5] illustrates that central regions of the brain are associated with trust in automation since central brain regions link to processes related to problem complexity, anxiety during attention tasks, and mental workload [21, 36, 66]. However, other previous studies [81] demonstrate that frontal and parietal regions can develop the most effective trust assessment models. These findings suggest that further research is necessary to draw definitive conclusions about whether a specific brain region alone can reliably detect trust calibration. The self-reported trust assessment model based on the whole brain region outperforms the models based on any single brain region, which is in line with prior work [81], but without presenting significant differences from the model that uses the central region only. This has important practical implications, as reducing the number of required EEG channels can simplify system implementation, lower computational complexity, and improve user comfort by minimizing the number of electrodes needed.

Our findings have important implications for system design, particularly in mitigating trust-calibration errors while users interact with system. For example, we can provide precise explanations through visuals to users, when users under-trust the system [56]. This might make the system more trustworthy to users. Besides, when users over-trust the system, we can provide a system confidence score or warning signals to users, so that users can calibrate trust properly [47, 61, 82]. Interactive AI interfaces should include mechanisms to collect and integrate small amounts of user-specific data during on-boarding or early interaction phases to personalize trust assessments effectively.

Despite these encouraging results, our work has several limitations. First, we did not incorporate time-domain EEG features into trust assessment models. Previous studies have indicated that time-domain features from central regions of the brain are associated with trust in AI [5], which can be investigated as part of our future work. Secondly, we did not incorporate a feedback system, which could inform users about their performance while they were going through the audio samples. This could potentially help users develop a better understanding of how the XAI works. Finally, EEG signals are influenced by numerous factors, such as environmental noise, physiological states (e.g., fatigue or stress) [16, 71], and individual variability [39]. These factors may confound the observed correlations between EEG features and trust calibration. For future work, we aim to study other neural measures with various degrees of temporal and spatial resolution.

8. Conclusion

We examined EEG power measures as proxies of self-reported trust in AI, and trust calibration. Our results indicate that Alpha, Delta, and Theta power waves correlate with self-reported trust, whereas Beta and Gamma power waves have the potential to differentiate properly calibrated trust against mis-calibrated trust. Machine learning models that estimate self-reported trust based on these measures achieve a Spearman's correlation ranging between ρ=0.21 to ρ=0.63 (p<0.01), depending on the setting of the model, and can further classify calibrated trust vs. mis-calibrated trust with F1 scores between 53-55%. Finally, our results suggest that the central brain region can provide trust assessment models with similar performance compared to all regions when estimating self-reported trust. These were discussed in terms of design implications for improving the interaction between human users and explainable AI models in decision-making.

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